Exposure-weighted versus lagged-treatment carryover specifications under differential-correlation biomarker interactions: a factorial simulation study

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Abstract

Background. Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Under the multivariate-normal (MVN) differential-correlation architecture (Architecture B), in which the biomarker-treatment interaction is encoded in a treatment-state-dependent correlation, a companion study found that carryover, that is, residual drug effect that decays after discontinuation, costs up to roughly 31% of power in relative terms. The analyst must still decide how to represent that residual exposure in the linear predictor. Unfortunately, it is not established which representation is most robust to the ordinarily unknown pharmacokinetic decay shape.

Methods. We compare two analysis-model specifications for the drug-exposure regressor under Architecture B: an exposure-weighted continuous indicator that assumes a parametric decay and half-life (M2, the current default), and the binary on-drug indicator augmented by a lagged just-off-drug term that estimates carryover non-parametrically (M3, the classical crossover device). We embed them in an ADEMP-structured factorial simulation[1] crossing two data-generating process (DGP) decay forms (exponential, Weibull) with the two specifications, under three trial designs, two sample sizes, and three interaction strengths (216 cells, 500 replicates each), together with four single-axis sensitivity blocks and a cluster-robust recalibration. We report every performance measure with its Monte Carlo standard error (MCSE).

Results. M2 attains the highest power only in the two designs with a blinded-discontinuation phase (Hybrid and OL+BDC), where its advantage over M3 is about 10 percentage points (0.860 vs 0.770 at the

reference cell). Conversely, under the classical crossover (CO) design M2 is markedly inferior (0.488 vs 0.830 for M3, which tracks the binary indicator closely). Where M2 leads, the lead appears robust to decay-shape and half-life mis-specification and to autocorrelation strength, and a paired-McNemar high-precision rerun confirms it at the highest-leverage cell (+7.2 points, $p = 1.2 \times 10^{-7}$), though the two specifications are statistically tied at three of four other rerun cells. The model-based interaction test is mildly conservative (mean null rejection 0.039), an artifact of the working covariance that a CR2 cluster-robust standard error removes without disturbing the ranking.

Conclusions. Three points warrant emphasis. First, under Architecture B we recommend the exposure-weighted predictor M2, reported with a cluster-robust standard error, for the Hybrid and OL+BDC designs; the lagged-treatment specification M3 is preferable, or at least not worse, under a crossover design. Second, across all conditions anticipated dropout governs power far more than the choice between the two specifications, and should dominate design effort accordingly. Third, we do not claim to have settled the wider comparison: this manuscript restricts the grid to Architecture B and to the exponential and Weibull decay forms, and does not evaluate the binary on-drug specification (M1) or Architecture A (mean moderation); the full-scope evaluation is retained in this repository as the archived companion drafts `archive/report.Rmd` and `archive/report-slim.Rmd`.

1 Introduction

In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator over a sequence of measurement occasions, and the within-patient contrast between the on-drug and off-drug periods estimates that patient's individual treatment effect. An aggregated N-of-1 trial pools many such single-patient series into a mixed model, trading some of the resolution available within a single patient for population-level inference and for sample sizes considerably smaller than a parallel-group trial would require[2]. The estimand of interest is the biomarker-treatment interaction, that is, whether a baseline covariate B_i modifies the treatment effect. We examine three trial designs that differ in how the on-drug and off-drug occasions are arranged, a classical crossover (CO), an open-label design followed by blinded discontinuation (OL+BDC), and a Hybrid N-of-1 design; each is defined in Section 2.

The inferential complication specific to this setting is carryover. When a drug is discontinued, its pharmacological effect does not switch off abruptly but decays over days or weeks, so that observations recorded during the nominally off-drug period still carry residual exposure. Treating those observations as though they were cleanly off-drug mis-specifies the design matrix, and the consequences for the interaction test are worth setting out explicitly.

The exposure regressor the analyst writes down is, in that case, a mis-measured version of the exposure the patient actually received, and the damage is of two kinds. First, the contrast that the interaction term is meant to estimate is diluted. Occasions that still carry a substantial fraction of the drug effect are entered with the same value as occasions that carry none, so the coefficient on the biomarker-by-exposure product is biased toward zero, in the familiar manner of a covariate measured with error. Second, those same occasions are heterogeneous in a way the model does not represent, and the unexplained variation is absorbed into the residual variance, inflating the standard error of the very coefficient that has just been attenuated. The test statistic loses on both counts, its numerator shrinking while its denominator grows, and power falls accordingly.

We note that this is a loss of power and not a loss of validity. Under the null, $c_{bm} = 0$, Architecture B generates no biomarker-response association at any level of residual exposure, so a mis-coded exposure regressor has no interaction to distort, and the size of the test is unaffected by the choice of exposure coding. The mild conservatism we do observe in Section 3.5 has a separate origin in the working covariance and applies equally to both specifications. Carryover mis-specification is therefore a question of efficiency rather than a source of spurious findings, and it is on that basis that we frame what follows in terms of recovered power.

But how should residual exposure be represented in the linear predictor? Two answers are in common use, and the question we take up here is which of them recovers more of the power that carryover would otherwise cost.

A companion study[3], building on Hendrickson[2], shows that under what we call Architecture B, a multivariate-normal differential-correlation formulation in which the interaction lives in the treatment-state-dependent covariance $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, carryover erodes power directly, and the loss is strongly design-dependent: at a half-life of one week, the relative loss is roughly 31% under OL+BDC, but only 3 to 5% under the crossover and Hybrid designs. Under an alternative architecture, in which the biomarker instead moderates the mean response directly, the corresponding loss is only 1 to 3% in every design; that architecture, together with a third combined architecture, falls outside the scope of the present manuscript, which restricts attention to Architecture B, where the carryover problem is largest and where Hendrickson's original results were obtained. Unfortunately, that earlier work fixed the decay shape at exponential and evaluated only a single analysis specification, leaving open the question of which representation of residual exposure performs best when the true decay shape is not known with any precision, as is ordinarily the case in practice.

We compare two ways of representing drug exposure in the linear predictor: an exposure-weighted continuous indicator that commits to a parametric decay form and half-life, which we call M2 and which is the current default in the pmsimstats-ng package, and the binary on-drug indicator augmented by a lagged just-off-drug term that estimates the magnitude of carryover without assuming any particular functional form, which we call M3, following the classical crossover-trial device described by Jones and Kenward[4]. We cross these two specifications against exponential and Weibull data-generating decay forms and ask whether one specification can be recommended outright, or whether the choice depends on the trial design. The study follows the ADEMP reporting framework of Morris, White, and Crowther[1], and every performance measure we report carries its Monte Carlo standard error. Section 2 describes the simulation design in ADEMP order, Section 3 gives the results, and Section 4 takes up the practical implications for trialists.

2 Simulation design

2.1 Aims

The aim of this study is to quantify, under Architecture B, the cost of mis-specifying the carryover representation in the analysis model, and to determine which of the two candidate strategies, M2 or M3, performs better across the grid of data-generating decay forms and trial designs. A second and related aim, addressed in the Tier 2 sensitivity blocks of Section 3.6, is to check whether the resulting ranking survives perturbation of the within-subject autocorrelation, the dropout rate, the accuracy of the assumed half-life, and the size of the biomarker-treatment interaction itself.

2.2 Data-generating mechanism

Under Architecture B, the biomarker and the biological response are drawn jointly from a multivariate normal distribution with a treatment-state-dependent correlation, $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ denotes the decay form and t_{sd} is the time elapsed since drug discontinuation. Architecture A, in which the biomarker instead moderates the mean response directly, and a combined Architecture C, fall outside the scope of the present manuscript; the full comparison across architectures is reported in the archived full-scope drafts (`archive/report.Rmd`, `archive/report-slim.Rmd`). In both architectures, the decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps t_{sd} to a residual-effect multiplier. We evaluate two such forms here, each parameterized to match a common half-life $t_{1/2}$. A third form, linear decay, appears in the full-scope manuscripts but is not considered further here.

2.2.1 Exponential decay

$$\phi_{\exp}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}}$$

This is the decay form used throughout the companion manuscript, and it is the one currently implemented in `implementations/tidyverse/R/functions.R`. It corresponds to classical first-order elimination kinetics and is the default assumption in pharmacokinetics; we retain it here as the reference against which the Weibull form is compared.

2.2.2 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

A shape parameter k governs whether the elimination tail is heavier than exponential ($k < 1$, slower elimination) or lighter ($k > 1$, faster off-drug clearance); the exponential form is recovered exactly at $k = 1$. We report results at $k = 0.7$, representing a drug with a prolonged low-concentration tail, and at $k = 1.5$, representing a drug cleared more rapidly once concentration falls below some saturation threshold, perhaps through active transport. Together with $k = 1.0$, which is numerically indistinguishable from the exponential form, this gives four decay-shape levels in the grid.

2.3 Estimand and analysis specifications

Under Architecture B, the interaction estimand is the on-drug biomarker-response correlation c_{bm}, a dimensionless quantity. Bias and empirical standard error are reported on a calibrated effect-size scale, namely the expected difference in on-drug response per standard deviation of the biomarker at $t_{1/2} = 0$, as defined in `docs/19-mean-moderation-implementation-notes.tex`; power and Type I error are reported on the native test-statistic scale. The null estimand used for Type I error verification is $c_{bm} = 0$, realized by the corresponding row of the parameter grid.

The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for both specifications considered here; they differ only in how the drug-exposure predictor X is constructed.

2.3.1 M2: exposure-weighted (continuous predictor)

Under M2, $X = \text{Dbc}_{it}$, a continuous exposure-decayed predictor constructed to match the analyst's assumed decay form and half-life; this is the current pmsimstats-ng default. At on-drug timepoints $D_{bc,it} = 1$ regardless of the assumed half-life. At off-drug timepoints the predictor decays as $D_{bc,it} = \exp(-(\ln 2/\hat{t}_{1/2})\, t_{sd,it}) = (1/2)^{t_{sd,it}/\hat{t}_{1/2}}$, losing exactly half its value for every $\hat{t}_{1/2}$ weeks that elapse after discontinuation. When the analyst's assumed decay form matches the true data-generating form, the predictor is well matched; when it does not, it is mis-specified in a way we probe directly in Tier 2 block S2. We note that the half-life assumption enters only through the predictor values handed to `lme`, not through the model formula, which is identical regardless of $\hat{t}_{1/2}$. That is, M2 is in effect a different model for every value of the assumed half-life, even though it is written the same way each time.

2.3.2 M3: lagged-treatment (estimated carryover term)

Under M3, $X = D_{it}$ is augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$, which marks the off-drug timepoints immediately following an on-drug timepoint. The magnitude of carryover is estimated from the data rather than assumed in advance, so this specification does not require the analyst to posit any particular decay form. The device follows the classical crossover-trial tradition surveyed by Jones and Kenward[4]. One structural feature of M3 bears directly on how the results below should be read: the lagged term L_{it} enters the model only as a nuisance main effect, and the interaction actually tested under M3 remains bm:D_{it}, the same coefficient tested under the binary on-drug specification (which we call M1 and do not evaluate further in this manuscript). M3 is best understood, then, as M1 with a carryover nuisance control added, rather than as a fundamentally different model of the interaction. Because it carries no half-life parameter, M3 is invariant to the analyst's assumed half-life by construction, in contrast to M2. This asymmetry, one specification indexed continuously by $\hat{t}_{1/2}$ and the other not a member of that family at all, is what the Tier 2 block S2 sensitivity contrast in Section 3.6 is designed to probe.

2.4 Factorial grid and parameter settings

The principal comparison is a factorial of DGP decay form (four levels: exponential; Weibull at $k \in \{0.7, 1.0, 1.5\}$) against analysis-model specification (M2, M3), under Architecture B only:

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks

Parameter	Values
Weibull shape (k)	0.7, 1.0, 1.5 (Weibull form only)
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	B (MVN differential correlation) only
Replicates per cell	500 (target); 50 for development

The 216-cell count follows the same nesting logic used in the full-scope manuscript, with the architecture dimension fixed at B and the linear decay form removed. The decay-shape combinations number four (exponential, and Weibull at $k \in \{0.7, 1.0, 1.5\}$), so the grid comes to $4 \times 3 \times 3 \times 2 \times 3 = 216$ cells, the factors being decay shape, carryover half-life, trial design, sample size, and biomarker level. The $c_{bm} = 0$ row serves as the Type I error check, targeting the nominal 5% level, and the $t_{1/2} = 0$ row, in which carryover is absent altogether, gives a best-case power reference at each combination of sample size, design, and c_{bm}, against which the power cost of nonzero carryover can be read off directly.

2.5 Performance measures and computing environment

For each cell we report Power, Type I error (at $c_{bm} = 0$), Bias and Empirical SE of the interaction estimate (on the calibrated scale), Coverage of the nominal 95% Wald CI, MSE, the non-convergence rate, and the MCSE of each measure, following Morris et al.[1, Tables 5-6].

We chose $n_{sim} = 500$ so that the Monte Carlo standard error on power at $p = 0.5$ would not exceed 0.022 ($\sqrt{0.5 \cdot 0.5/500} \approx 0.022$). No simulation is run for this manuscript; we instead filter the same production-scale grid summary used in the archived full-scope drafts (540 cells, `analysis/data/02-grid-summary.rds`, computed with `implementations/tidyverse/` as the primary implementation under R 4.5.3, 128 minutes on eight cores, commit SHA `7018b59`, dated 2026-04-15) down to the 216-cell slice defined by Architecture B, the two non-linear decay forms, and the M2 and M3 specifications. Within each cell, both remaining specifications were fit to the same

Table 1: Principal simulation grid (Architecture B only). Cell 1 is the matched reference: exponential DGP analyzed under the current pmsimstats-ng default. Off-diagonal cells quantify the cost of DGP-analysis mis-specification.

DGP decay	M2 exposure-weighted	M3 lagged-treatment
Exponential	cell 1 (matched)	cell 2
Weibull (k = 0.7)	cell 3	cell 4
Weibull (k = 1.0)	cell 5	cell 6
Weibull (k = 1.5)	cell 7	cell 8

simulated dataset (common random numbers), so the M2 minus M3 contrast is estimated with reduced Monte Carlo variance relative to what independent draws would give. The Tier 2 sensitivity blocks and the S6 cluster-robust recalibration are filtered from the same shared outputs (`analysis/data/02-sensitivity-summary.rds` and `analysis/scripts/carryover-sensitivity/output/diag-s6-cr2.rds`); their designs were already anchored at Architecture B. The filtered figures used here are produced by `03-render-figures-extra-slim.R`, `04-sensitivity-figures-extra-slim.R`, and `07-type1-cr2-figure-extra-slim.R`.

3 Results

All results reported below derive from the same 500-replicate production run used in the full-scope manuscripts, filtered to Architecture B, the exponential and Weibull decay forms, and the M2 and M3 specifications. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$, and differences smaller than this should not be over-interpreted. Coverage of the nominal 95% Wald confidence interval is reported for the reference cell, and non-convergence rates are reported per cell in the tables that follow.

3.1 Headline performance table

Table 2 reports performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$), where non-convergence was zero throughout. A caveat applies to the bias, mean squared error, and coverage columns: these are computed against M2's calibrated target, $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$, while M3 estimates a

Table 2: Headline performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). MCSEs are reported per Morris et al. (2019, Table 6).

t1/2	spec	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	M2	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	M3	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	M2	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
0.5	M3	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
1.0	M2	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000
1.0	M3	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000

different coefficient, bm:D_{it}. M3's apparent bias and under-coverage in this table reflect that mismatch in estimand rather than any defect in the estimator, and for that reason we compare only power and Type I error across the two specifications; the bias and coverage columns are reported for completeness but are not directly comparable between M2 and M3.

3.2 Matched-decay baseline

At the matched cell (exponential DGP, M2 exposure-weighted analysis, Architecture B, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate plus or minus MCSE, $n_{sim} = 500$). All 500 replicates converged at every half-life in this cell. The value at $t_{1/2} = 1.0$ agrees with Hendrickson's reported power of 0.82 for the same cell[2], which we take as reassurance that the simulation pipeline reproduces the baseline result correctly.

3.3 Power by analysis specification

The two specifications give nearly identical power at $t_{1/2} = 0$, where there is no carryover to model, and diverge progressively as the half-life increases. M2 retains a modest but consistent advantage in the Hybrid and OL+BDC designs, an advantage that first appears at $t_{1/2} = 0.5$ weeks and grows through $t_{1/2} = 1.0$ weeks. This advantage does not appear under the crossover design, where M2 is in fact dominated, a point we return to in Section 4. The mechanism appears straightforward: in a crossover trial, the within-subject correlation already carries most of

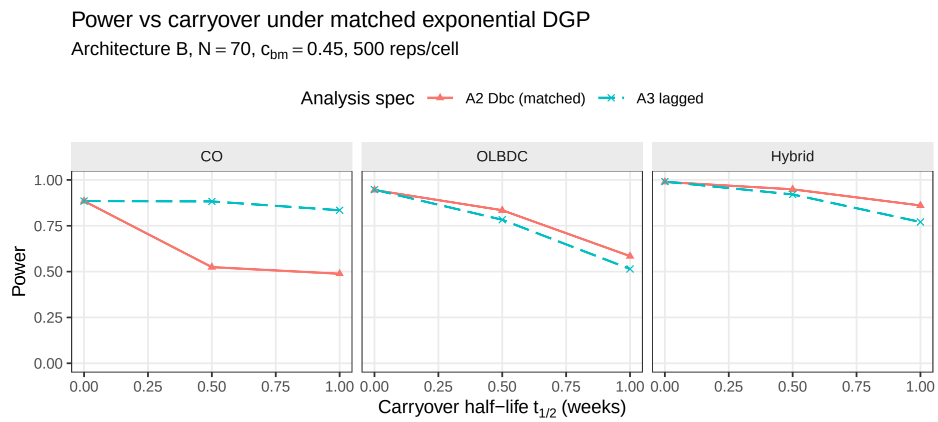

Figure 1: Power versus carryover half-life under the exponential DGP, Architecture B, stratified by analysis specification. M2 retains a modest advantage over M3 in the Hybrid and OL+BDC designs at $t_{1/2} = 1.0$ weeks, but is dominated by M3 under the crossover (CO) design.

the signal that M2's decaying predictor was designed to recover, so M2 has little left to gain, and its down-weighting of the off-drug observations only works against it.

3.4 Decay-form mis-specification

The heatmap cell at the intersection of the exponential DGP row and the M2 column is the matched reference, with power 0.860 ± 0.016. The same row's M3 column gives the power penalty of the lagged specification when the data-generating decay is exponential, while the other rows of the M2 column give the cost of assuming exponential decay in the analysis when the true decay form is Weibull. Even under the heaviest-tailed Weibull data-generating process we examined ($k = 0.7$), M2's advantage over M3 narrows but does not reverse. We take up the practical interpretation of this in Section 4.

3.5 Type I error control and cluster-robust recalibration

Across the null cells, empirical Type I error sits uniformly below the nominal 0.05 level for both specifications and in all three designs. This shortfall is not sampling noise but a genuine conservatism of the model-based interaction test, which the companion calibration study[5] traces to the corCAR1 working covariance: the estimated standard error runs six to

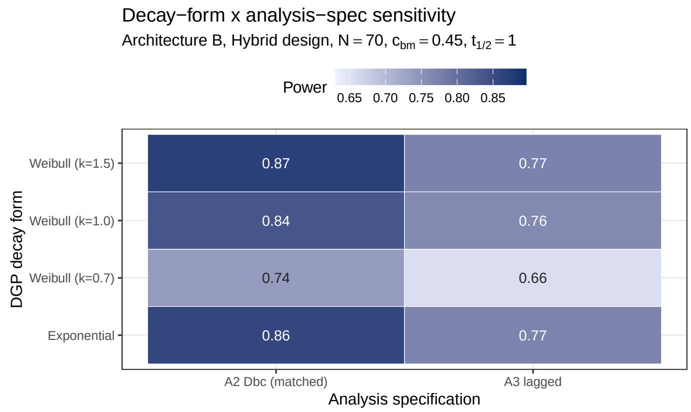

Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the heavy-tailed Weibull DGP ($k = 0.7$), M2's advantage over M3 narrows because its assumed exponential decay no longer matches truth. The fill scale is stretched to the observed power range (0.66-0.87), not the full [0,1] range, so that these cell-to-cell differences are visible; the absolute shading should not be compared to other figures in this manuscript.

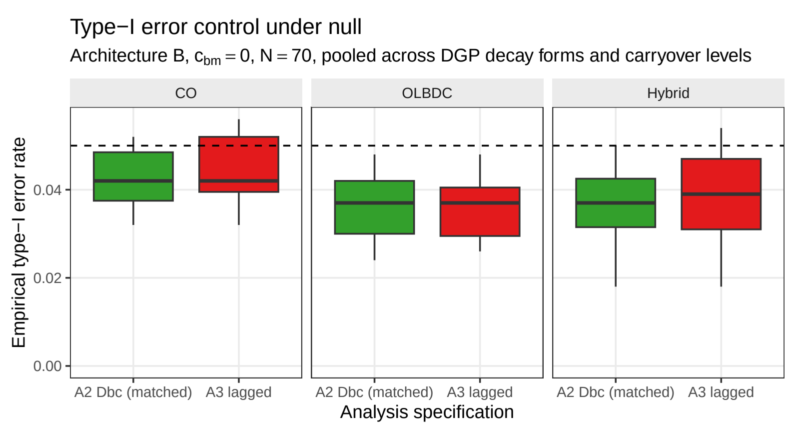

Figure 3: Empirical Type I error rates under the null ($c_{bm} = 0$), Architecture B, pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

Table 3: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, Architecture B, 3,000 replicates). κ is the ratio of the mean standard error to the empirical standard deviation of the interaction estimate; $\kappa > 1$ indicates a conservative test. Both specifications are conservative under the model-based test, most so the exposure-weighted M2 ($\kappa = 1.10$, Type I = 0.029); the CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for both specifications.

Specification	κ model	Type I model	κ CR2	Type I CR2
M2	1.10	0.029	1.00	0.045
M3	1.05	0.039	0.98	0.049

Table 4: Cluster-robust recalibration of M2 (Power columns) and of the M2−M3 ranking (last two columns), at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$, Architecture B). κ is the ratio of the mean standard error to the empirical standard deviation of the interaction estimate for M2; a value above one is a conservative test. The M2−M3 columns are computed directly from the per-specification power at this cell under each inference, so they are a genuine check of this manuscript's own comparison rather than the M2-vs-M1 contrast reported for the full-scope grid. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test on M2. Computed with `clubSandwich`.

Cell	κ model	κ CR2	Power model	Power CR2	M2−M3 model	M2−M3 CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.082	+0.090	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.068	+0.086	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.036	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.012	+0.022	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.014	-0.000	5

ten percent too large, and the realized rejection rate falls correspondingly. Because the conservatism is uniform across specifications, it does not disturb the M2-versus-M3 comparison, and the cluster-robust recalibration reported below (block S6) confirms that a CR2 cluster-robust standard error restores the rejection rate to nominal without changing which specification leads. Table 3 makes the per-specification conservatism, and its cluster-robust correction, explicit at the null cell.

Figure 4 shows the effect of recalibration on test size directly. The model-based null rejection rate sits below nominal across all five S6 cells and both specifications, while the CR2 test restores it to within the Monte Carlo band around 0.05. This matters because a test that is not correctly sized cannot fairly be preferred on power alone. In the information-rich

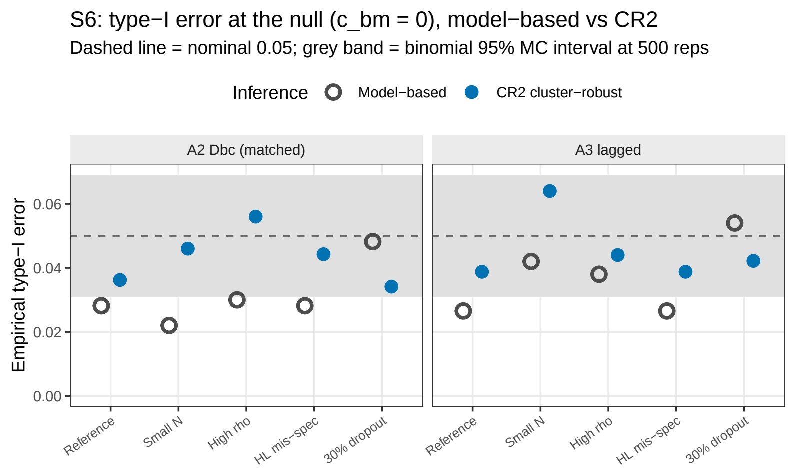

Figure 4: Empirical Type I error at the null ($c_{bm} = 0$) under model-based and CR2 cluster-robust inference, across the five S6 stress cells and the two specifications (Architecture B). The model-based test (open) is conservative (near 0.03); CR2 (filled) is restored to the nominal 0.05 (dashed), within the binomial Monte Carlo band (gray). 500 null replicates per cell.

cells, the calibration ratio κ runs from 1.07 to 1.26 under the model-based test and is restored to about one under CR2, and the correctly sized test recovers two to five points of power in the process. Table 4 reports the M2 minus M3 gap, which is this manuscript's own comparison, computed directly from the per-specification power at each S6 cell rather than borrowed from the M2-versus-M1 contrast reported in the full-scope grid. That gap is held or slightly widened under CR2 at the reference cell (+0.080 to +0.092), at the small-N cell (+0.068 to +0.086), and at the half-life mis-specification cell (+0.012 to +0.022); it narrows somewhat under high autocorrelation (+0.052 to +0.036, though M2 still leads); and at 30% MCAR dropout it collapses from a small M2 lead under the model-based test (+0.014) to an essentially null difference under CR2 (−0.0002), which we read as a statistical tie. We do not regard this as a contradiction of the ranking so much as a restatement, from a different angle, of the dropout finding in block S3 below: M2's advantage is driven by information carried in the off-drug observations, and once dropout has removed most of that information there is not much left for either the model-based test or its cluster-robust correction to distinguish. This recalibration was validated only in the Architecture B, Hybrid, $N = 70$ cells, and it should not be extended to the crossover design, where M2 does not lead in the first place.

3.6 Sensitivity analyses

Four Tier 2 blocks perturb one axis at a time around a common reference cell (Architecture B, Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$), holding the remaining factors fixed. The blocks are reported separately here and are not pooled with Tier 1 or with each other.

3.6.1 S1: within-factor autocorrelation

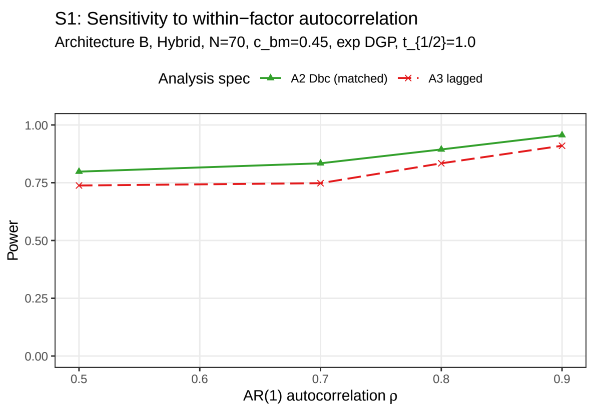

Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$, Architecture B.

Both specifications gain power monotonically with ρ. At $\rho = 0.5$, M2 power is 0.798 ± 0.018 against M3's 0.738 ± 0.020; at $\rho = 0.9$, M2 reaches 0.956 ± 0.009 against M3's 0.910 ± 0.013 (point estimate plus or minus MCSE, $n_{sim} = 500$). The M2 advantage over M3, roughly 0.06 in absolute power throughout, is preserved at every autocorrelation level we examined, which suggests that autocorrelation strength and the choice of specification act additively rather than interacting with one another.

3.6.2 S2: analyst-truth half-life mismatch

M2 appears fairly robust to a mismatch between the analyst's assumed half-life and the true data-generating half-life. Across the eight S2 cells (DGP $t_{1/2} \in \{0.5, 1.0\}$ weeks crossed with analyst-assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), M2 averages 0.868, ranging from 0.766 to 0.964, with per-cell MCSE between 0.008 and 0.019. M3, which does not consume the half-life parameter, is invariant to the analyst's assumption by construction, and its power averages 0.840 across the same eight cells. The exposure-weighted predictor's curvature is gentle

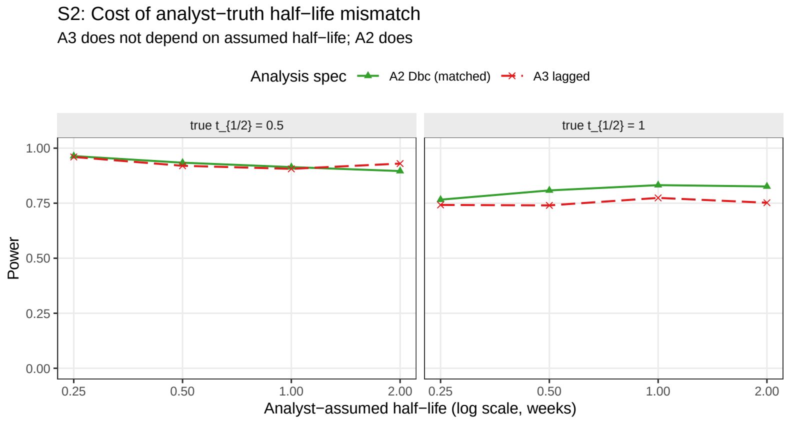

Figure 6: Power of M2 and M3 as the analyst's assumed half-life departs from the true DGP half-life. M3 is invariant to the assumed half-life by construction; M2's power degrades with mis-specification.

enough that mis-specifying the assumed half-life by a factor of two does not erode M2's advantage over M3 at either true half-life. It would appear, then, that M2 remains a defensible choice even when the true pharmacokinetic half-life is known only approximately at the design stage.

3.6.3 S3: dropout

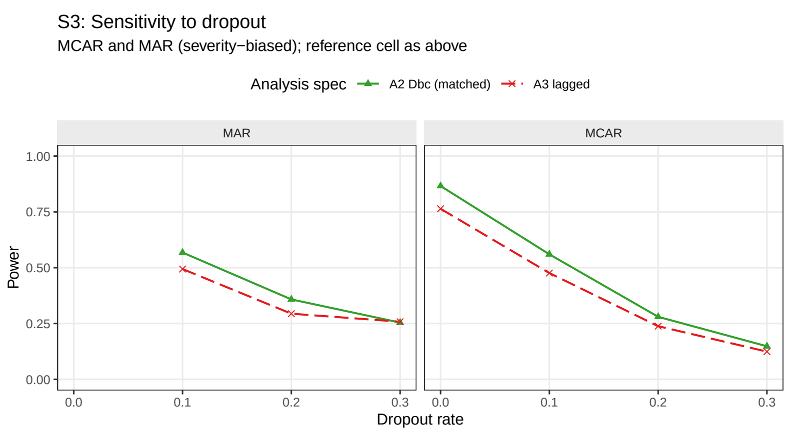

Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms, Architecture B.

Both specifications lose power monotonically as the dropout rate increases. At the MCAR baseline ($N = 70$, no dropout), M2 gives 0.866 ± 0.015 and M3 gives 0.764 ± 0.019; at an MCAR dropout rate of 0.30, these collapse to 0.148 ± 0.016 and 0.124 ± 0.015 respectively. M2's advantage narrows steadily as dropout worsens: the absolute M2 minus M3 gap runs about 0.10 at zero dropout, 0.07 at a rate of 0.10, 0.04 at 0.20, and 0.03 at 0.30 under MCAR, narrowing further under MAR dropout at the highest rate examined (about 0.004 at rate 0.30). This pattern is consistent with M2's advantage originating in the exposure-weighted contrast contributed by off-drug observations; when those observations are preferentially lost, M2's edge contracts toward M3's. At low to moderate dropout, that is, at a rate of 0.20 or below, the M2 advantage is preserved.

3.6.4 S4: biomarker-moderation effect-size curve

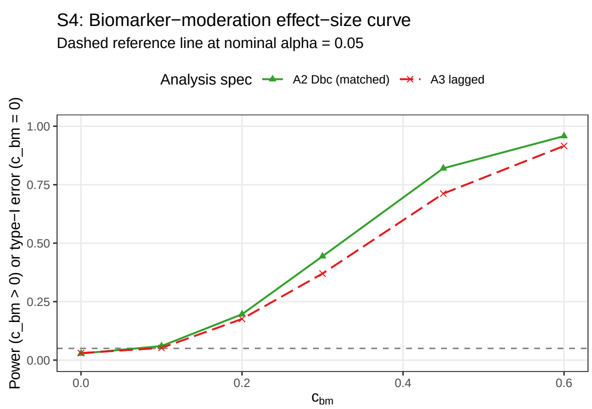

Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$ under Architecture B. The $c_{bm} = 0$ point is the null for Type I verification.

The power curve is sigmoidal for both specifications, as expected. At $c_{bm} = 0$, empirical Type I error sits below nominal, at 0.028 ± 0.007 for M2 and 0.030 ± 0.008 for M3; power then rises through the mid-range and saturates once $c_{bm} \geq 0.45$, reaching 0.958 ± 0.009 for M2 and 0.916 ± 0.012 for M3 at $c_{bm} = 0.6$. The M2 advantage is largest at intermediate effect sizes, $+0.082$ absolute at $c_{bm} = 0.30$ and $+0.102$ at $c_{bm} = 0.45$, shrinking to $+0.042$ at $c_{bm} = 0.60$ as both specifications approach the ceiling. The practical implication is that the advantage matters most in trials targeting a moderate biomarker-treatment interaction, and matters least when the true effect is either very small or comfortably large.

3.7 A higher-precision check

Because a number of the Tier 1 gaps are small, we reran a focused 24-cell grid (OL+BDC design, $N \in \{35, 70\}$, true half-life $\in \{0.5, 1.0\}$) at

600 trials per cell, with all trials converging. Because M2 and M3 are fit to the same trials, we can compare them with a paired McNemar test, which is considerably more sensitive than treating the two specifications as independent groups. Type I error across the twelve null cells of the rerun fell in $[0.003, 0.053]$, so both specifications hold nominal alpha at the sample sizes and half-lives examined. Paired McNemar tests on M2 versus M3 find M2 strictly ahead only at the highest-leverage cell ($N = 70$, true $t_{1/2} = 1.0$ weeks, $\Delta = +7.2$ percentage points, $p = 1.2 \times 10^{-7}$); in the remaining three cells the paired difference lies between -0.5 and $+1.5$ percentage points, with $p \geq 0.18$, and the two specifications are operationally indistinguishable at this level of precision. Resolving the M2-versus-M3 gap at the intermediate cells with statistical clarity would require on the order of 2,000 replicates per cell, given the current Monte Carlo standard error.

4 Discussion

4.1 The principal finding, and the conditions under which it fails

The principal finding is that M2's advantage over M3 is real, but conditional. On the one hand, at its best cell (Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$), M2 exceeds M3 by about 10 percentage points of power (0.860 versus 0.770), a difference of roughly five Monte Carlo standard errors and therefore not attributable to chance. On the other hand, under the crossover design M2 performs considerably worse (0.488 versus 0.830 for M3). The reason for this reversal appears straightforward: in a crossover trial, the within-subject correlation already carries most of the signal that M2's decaying predictor was designed to recover, so M2 gains little from its more elaborate representation of exposure, and its down-weighting of off-drug observations works against it. The recommendation that M2 is to be preferred therefore holds only in the two non-crossover designs studied here, and should not be extended to a crossover trial.

4.2 What the robustness checks add

Three checks probe whether M2's advantage, where it exists, is fragile. On decay shape (Figure 2): even when the analyst incorrectly assumes exponential decay and the true form is Weibull, M2 still does not fall below M3, within the range of decay shapes we examined. On the assumed half-life (Figure 6): M3 does not use a half-life and is therefore unaffected, while M2 does, yet mis-specifying the half-life by a factor of two or four moves M2's power only from about 0.83 down to 0.77, never below M3. On autocorrelation (Figure 5): stronger serial correlation raises both specifications by roughly the same amount, so M2's lead of five to six points holds regardless of correlation strength. Taken together, these checks suggest that where M2's advantage exists, it is not a fragile artifact of one particular combination of nuisance parameters.

4.3 Dropout matters more than the method

The finding with the largest practical consequence in this study concerns missing data, not carryover. Unfortunately, when patients drop out (Figure 7), power collapses for both specifications together: at 30% dropout, both fall to somewhere between 0.12 and 0.25 and become essentially indistinguishable. M2 offers no protection against this loss; it can only organize whatever data happen to remain. Dropout tied to symptom severity (MAR) is somewhat worse than dropout occurring at random (MCAR), because the sickest patients, who carry the most signal, tend to leave first. The practical implication is direct: retaining patients in the trial yields considerably more statistical power than the choice between M2 and M3 ever could.

4.4 Scope, related work, and limits

This manuscript deliberately narrows the full-scope grid to Architecture B, to the non-linear decay forms, and to the M2-versus-M3 contrast, in the interest of a self-contained evaluation of manageable length. The archived full-scope drafts (`archive/report.Rmd` and `archive/report-slim.Rmd`) show that the M2 advantage documented here does not carry over to Architecture A, where the biomarker moderates the mean response directly and M2 is marginally dominated in every design we examined; nor is the binary on-drug specification M1 dominated by M2 outside the Hybrid and OL+BDC cells under Architecture B considered here. Readers who need the cross-architecture comparison, or the comparison against M1, should consult that manuscript rather than extrapolate from this one.

Within the narrower scope adopted here, our matched-decay power at the headline cell, 0.860 ± 0.016, agrees with Hendrickson's[2] reported 0.82 for the same cell. The crossover tradition surveyed by Jones and Kenward[4] would lead one to expect that a shape-free lagged term such as M3 should outperform a parametric predictor such as M2 once the decay shape is mis-specified. Our S2 result does not support that expectation over the range we examined, because M2's penalty for an incorrect half-life appears too small for M3 to overtake it.

Two further limitations bound what we can conclude. First, the numbers presented here come from a single parameter set, calibrated to a prazosin and post-traumatic stress disorder trajectory, so the absolute power values should be regarded as illustrative rather than general; the ranking is the more portable part of the result, and even the ranking is conditional on the trial design, as we have shown above. Second, the sensitivity checks vary one factor at a time and can therefore miss genuine interactions between factors, for instance between high autocorrelation and heavy dropout; the one joint check available to us, reported in the full-scope manuscript as block S5, revealed no such interaction.

5 Conclusions

1. **M2 helps, but conditionally.** The exposure-weighted predictor achieves the highest power, about 10 points over M3, only in the Hybrid and OL+BDC designs, and its lead survives an incorrectly assumed decay shape and an incorrectly assumed half-life within those designs. Under the crossover design, M2 confers no advantage, and M3 is at least as good. Where M2 is used, we recommend reporting it with a CR2 cluster-robust standard error (Block S6).

2. **An approximate half-life is adequate, where M2 is used.** Mis-specifying the half-life by a factor of two to four moves M2's power only modestly, so a rough pharmacokinetic estimate is sufficient in practice.

3. **Dropout matters more than either method.** Power falls sharply as dropout increases, roughly halving at a dropout rate of 20%, and this loss affects both specifications equally. Preventing dropout is a more effective use of design effort than choosing

between M2 and M3.

4. **The interaction test is somewhat conservative, and the conservatism can be corrected.** Both specifications reject slightly less often than the nominal 5% under the null, and a CR2 cluster-robust standard error restores the correct rate while recovering a modest amount of power. The M2-versus-M3 gap is held or widened under CR2 at four of the five S6 stress cells, but collapses to a statistical tie under 30% dropout, consistent with the third conclusion above (Block S6).

5. **This is a narrowed slice of a larger study.** Architecture A, the binary specification M1, and linear decay all fall outside the scope of this manuscript; the full-scope grid reported in the archived drafts `archive/report.Rmd` and `archive/report-slim.Rmd` is the authority on those comparisons, and should be consulted before generalizing beyond Architecture B.

6 Conflict of interest

The authors declare no conflicts of interest.

7 Funding

This work received no specific funding.

8 Data availability statement

The data and code supporting the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths for this manuscript are listed in the Reproducibility section below.

9 Reproducibility

This manuscript was rendered with R 4.6.0 inside the project's zzcollab Docker container; the production simulation itself ran under R 4.5.3, as recorded in §3.6, with the image hash recorded in `Dockerfile` and the package set captured in `renv.lock`. Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. No simulation code runs during knitting; the manuscript loads pre-computed checkpoints only.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds are derived from the master seed as `master + rep_idx`.

Data and code availability. Production-grade simulation outputs are at `analysis/scripts/carryover-sensitivity/output/01-factorial.rds` (principal factorial) and `04-sensitivity.rds` (sensitivity blocks S1 through S5). Cell-level summaries are at `analysis/data/02-grid-summary.rds` and `analysis/data/02-sensitivity-summary.rds`. Driver scripts are `01-run-factorial.R`, `04-run-sensitivity-blocks.R`, and `02-summarise-grid.R` under `analysis/scripts/carryover-sensitivity/`. The Architecture B filtered figures used in this manuscript are produced by `03-render-figures-extra-slim.R`, `04-sensitivity-figures-extra-slim.R`, and `07-type1-cr2-figure-extra-slim.R` in the same directory. The simulation grid plan and design rationale are documented at `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md`, which serves as the Morris ADEMP pre-registration. High-precision prototype rerun outputs are at `analysis/data/quick-sim/02-carryover-replicates.rds` (per-replicate) and `02-carryover-summary.txt` (Morris ADEMP cell-level summary). The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/02-carryover-sensitivity/`; superseded wider-scope drafts are retained under `archive/` in that directory and are documented in its `README.md`.

Specification labels. The three specifications are recorded as `A1`, `A2`, `A3` in the archived simulation output and are reported here as M1, M2, M3; the mapping is applied at display time and the stored values are unchanged.

10 References

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